

Introduction to Segmentation

1. GOAL

Introduction to unsupervised methods for image object segmentation: pre-processing, binarization, post-processing
Introduction to image filtering (pre-processing), Otsu thresholding (binarization), mathematical morphology (post-processing)
Influence of kernel parameters in filtering and morphological operations

2. MATERIALS:

Session slides, python code (PyCode_BasicSegmentation.py) and annotated volumes (VOIs.zip, MetadatabyNoduleMaxVoting.xlsx).

The file VOIs.zip contains two folders:

- *Image*: contains, for each nodule, .nii.gz files of the VOIs of the CT scan named with the ID of the patient and the ID of the nodule (LIDC-IDRI-0003_R_2.nii.gz, for nodule 2 of patient LIDC-IDRI-0003)
- *nodule_mask*: contains .nii.gz files of a binary mask of each nodule VOI with the same name that identifies the nodule in the *image* folder

All materials are available at Campus Virtual.

3. SESSION EXERCICES

Exercise 1. Volume Binarization. Visualize Otsu binarization for the different Cases and lesions and analyze the quality of the segmented nodule compared to the original ROI intensity volume and the manual segmentation.

- a) What Otsu segmentations contain surrounding tissue? Discuss an explanation for that artifact.
- b) What Otsu segmentations have internal holes? Discuss an explanation for that artifact.

Exercise 2. Volume Post-Processing. Use mathematical morphology to:

- a) Separate lesions from surrounding tissue. What operation do you need? What structural element (size and shape) have you used? Does it depend on either the case or ROI size?
- b) Reduce nodule internal holes. What does the shape and size of the structural element depend on? Define a criteria to choose the size of the structural element.

Exercise 3. Volume Pre-Processing. Apply linear and non-linear filterings to reduce the impact of radiological noise in segmented lesions and compare segmentations for each type of image filtering:

- a) Apply a (linear) Gaussian filtering to volumes before Otsu thresholding and analyze visually the impact on segmentations. Compare lesion contours (boundary) for different scales.
- b) Apply a (non-linear) Median filtering to volumes before Otsu thresholding and analyze visually the impact on segmentations. Compare lesion contours (boundary) for different filter sizes.

Exercise 4. Segmentation Strategy. Define the best strategy to segment nodules. Justify your choice in terms of advantages and disadvantages under different conditions (radiological noise, lesion location, ROI size) in input data